



LUND
UNIVERSITY

Numerical Analysis for E3
Numerical Analysis, Centre for Mathematical Sciences

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Homework assignment 5

Delivery date: Tuesday, May 13th., not later than 18:00.

Topics of this assignment: nonlinear systems of equations; iterative methods; Newtons method; order of convergence.

1. A nonlinear equation. Consider the function $y(x)$ given by

$$y(x) = x - 4 \sin(2x) - 3.$$

Task 1a): Plot the function in order to get an idea where the zeros are.

Task 1b): Use Newton's method to compute the zeros. Terminate the iteration when $\left| \frac{\Delta x^{(n)}}{x^{(n)}} \right|$ is close to machine precision (see Hint). What are the roots? How many iterations do you need for full convergence?

Task 1c): Now consider the root $x^* = -0.54$. Plot the absolute errors $|x_{k+1} - x^*|$ against $|x_k - x^*|$ for several iteration steps k . Verify the order of convergence from this plot. (How can you read off the order of convergence from this plot?)

Hint: a "skeleton" for Newton's method in MATLAB could look like

```
function x=newton(x,tol)
dx=1;
iter=0;
while abs(dx/x)>tol & iter<9
    dx=f(x)/fprim(x);
    x=x-dx;
    iter=iter+1;
end
```

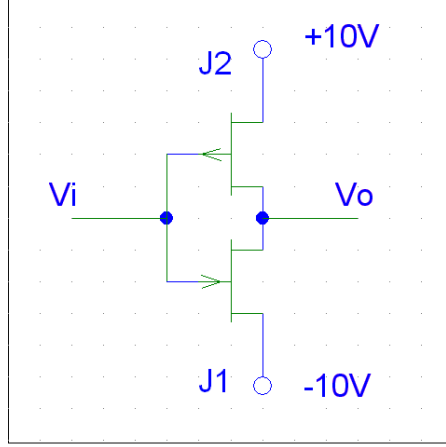


Figure 1: CMOS Digital inverter using MESFETs

2. Newton's method for systems. A Mesfet inverter. We consider the digital inverter given in Fig. 1. It is constructed by two MESFET transistors and is supposed to answer to a positive input voltage V_i by a negative output voltage V_o and vice versa.

For this MESFET inverter the following relation between V_i and V_o holds:

$$(2(V_i + 9)(V_o + 10) - (V_o + 10)^2) \left(\frac{6}{5} - \frac{1}{50} V_o \right) - (V_i - 9)^2 \left(\frac{6}{5} + \frac{1}{50} V_o \right) = 0 \quad (1)$$

This equation implicitly defines $V_o = V_o^-(V_i)$ as a function of V_i .

Next, we are interested in the so-called logic point of decision. It is defined as the point (V_o, V_i) for which $\frac{dV_o}{dV_i} = -1$. Taking an implicit derivative of (1) with respect to V_i , inserting $\frac{dV_o}{dV_i} = -1$ it follows:

$$\frac{117}{25} V_o - \frac{119}{25} V_i - \frac{1}{10} V_o^2 + \frac{1}{25} V_i V_o + \frac{2561}{50} + \frac{1}{50} V_i^2 = 0 \quad (2)$$

Your task is to calculate logic point of decision (V_o, V_i) which is defined by the system of non-linear equations (1) and (2). Determine the physically meaningful solution of the problem.

3. "Division free" division. Given a real number $A \neq 0$, the inverse $1/A$ is a root of the function $f(x) = 1/x - A$. Hence, one might apply Newton's method to f in order to approximate the inverse $1/A$.

Task 3a): Show that the method is given by

$$x_{n+1} = 2x_n - x_n A x_n.$$

Note that this iteration is "division free".

Task 3b): Apply the method to approximate π^{-1} .

Task 3c): Show that the method converges quadratically for $A \neq 0$.

Task 3d): Apply the method to approximate the inverse of the matrix:

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}; \quad \text{initial guess: } X_0 = \begin{pmatrix} -0.5 & 1 \\ 1 & -0.5 \end{pmatrix}.$$

4. Iterative improvement. A numerical approximation $\tilde{x}$ to the solution of a regular linear system $Ax = b$ can be interpreted as an approximate root of the vector-valued function

$$F(x) = Ax - b.$$

Iterative improvement is a method to improve the accuracy of the approximation. The idea is to use $\tilde{x}$ as an initial guess and to apply Newton's method to compute better approximations to the root of F . The algorithm is as follows: Let $x^{(k)} = \tilde{x}$ and iterate:

- Step 1: Compute the residual $r^{(k)} = Ax^{(k)} - b$.
- Step 2: Solve $A\Delta x^{(k)} = -r^{(k)}$.
- Step 3: Let $x^{(k+1)} = x^{(k)} + \Delta x^{(k)}$

Note that the Jacobian of $F(x)$ is $F'(x) = A$. Furthermore, to solve the system in Step 2 any earlier computed factorization of the matrix A (LU , Cholesky or QR) should be reused to reduce the complexity.

Iterative improvement is an example of defect correction, a powerful tool in scientific computing. The basic idea is to successively correct the defect (=residual) until it is acceptably small.

Task 4a): Assume that in the above algorithm all arithmetic operations are done exactly. Show that the iteration “converges” in only one step, i.e. $x^{(1)}$ is the exact solution of the system $Ax = b$.

In practice, however, computations are inexact. Therefore, when applying iterative improvement, it is recommended to compute the residual in Step 1 in higher accuracy. In MATLAB this may be done using the `vpa`¹ command:

```
r = vpa(A*x-b,64) ;
```

Task 4b): Apply iterative improvement to the Hilbert system in Homework 3, Exercise 3. Use Cholesky decomposition to solve the linear systems. Compute (not more than) 10 iteration steps.

¹To run the `vpa`-command, the MATLAB symbolic toolbox must be installed on your system. If you don't have it, you can't do part b) of this exercise.